



Introduction to Software Development

Software Development COS7009–B

Who is who

- Module Coordinator: Dr Ci Lei
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- Office: Chesham B3.08

- Research/Teaching Interests: Quantum Computing, and applications to AI, etc
- Web link: <https://www.bradford.ac.uk/staff/clei1>
- Availability: email to arrange

Tell us about you

- Your name
- The programme you are studying
- Your background
- Your plan

Module Organisation

- 20 credits [200 study hours]
- Lectures: 24 hours – Tuesdays 10:00–12:00
- Labs: 12 hours – Fridays 12:00–13:00
- Direct study & assessments: 164 hours
- Lecture materials available in CANVAS

Assessments

- **Two individual assessments:**
 - MCQ test (50%) (close-book, 2 hours):
 - During Exam Period*
 - All LOs will be assessed with a focus on the fundamentals of programming language.
 - Programming exercises (50%) (open-book, 2 hours):
 - During Exam Period*
 - All LOs will be assessed with a focus on problem analysis and solving.
- **Mock Exams will be held for your exam preparation**
- **All materials presented in lectures, labs and recommended readings are examinable**
- **Provisional marks and feedback date: 4 weeks after exams at the latest**

*<https://unibradford.ac.sharepoint.com/sites/exams-intranet/SitePages/Information-for-Students.aspx>

Extension and Extenuating Circumstances

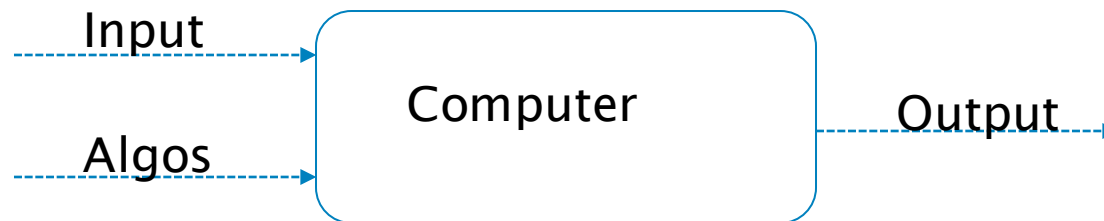
- Refer to University Process

<https://unibradfordac.sharepoint.com/sites/registry-and-student-admin-intranet/SitePages/Extenuating-Circumstances.aspx>

- What you should do if you need CW extension or need to apply for EC.
- Personal Tutors can also contribute to your CW extension and EC applications (if it is genuine).

What is Software Development

- Software and its development process have evolved over the years
- Software: A solution to a problem in the form of code
- Development: a whole spectrum of tasks involved, varies depending on the nature of the problem



What is this module about?

In this module, we focus on

- syntax,
- data structure,
- algorithms,
- paradigm

We use Python as the language

- Simple
- Best practices
- User base

References & resources

- Python programming for the absolute beginner By Dawson Michael
- The Practice of Computing using Python by Punch and Enbody
- Learning Python 3rd and 4th editions by Lutz, Mark
- Fundamentals of Python: data structure by Lambert, Kenneth Alfred
- Python Algorithms: Mastering basic algorithms in the Python language, Second edition By Hetland, Magnus Lie
- Practical Python By Hetland, Magnus Lie
- Python: high performance programming By Lanaro, Gabriele
- <http://faculty.washington.edu/stepp/>
- <https://www.py4e.com/lessons>

Any Questions